

Numerical Analysis PhD Exam, September 2000

Do any 8 of the following 10 problems.

Part 1: Numerical Linear Algebra

1. A square matrix A which can be decomposed into a form $A = SDS^{-1}$, where D is diagonal, is said to be diagonalizable.

(a) Is the matrix

$$A = \begin{pmatrix} 5 & 1 \\ 0 & 3 \end{pmatrix}$$

diagonalizable? Why?

(b) Can the above matrix A be diagonalized with a matrix S such that $S^*S = I$?

2. This problem has three parts.

- (a) State a procedure for factoring a matrix A (without pivoting) into a form $A = LDU$, where L is unit lower triangular, U is unit upper triangular, and D is diagonal.
- (b) State sufficient conditions on A for the procedure in part (a) (without pivoting) to work.
- (c) What is the computational advantage of having lower and upper triangular matrices? An explanation without specific computation counts is sufficient.

3. This problem has two parts.

- (a) Under what conditions can a matrix be factored into a form $A = U^*TU$, where $U^*U = I$, and T is upper triangular?
- (b) If A is Hermitian, how is U constructed? (You don't need to actually give an algorithm.)

4. This problem has three parts.

- (a) State the power method algorithm for finding a dominant eigenvalue and an associated eigenvector of a matrix A . (An eigenvalue λ_i is said to be dominant if $|\lambda_i| > |\lambda_k|$ for all other k .)
- (b) Show that the power method converges (for almost any starting value) when A is real symmetric and has a dominant eigenvalue.
- (c) Explain how the inverse power method can be used to find non-dominant eigenvalues and eigenvectors for a matrix A .

5. This problem has three parts.

- (a) Give a careful statement of the Gershgorin Circle Theorem.
- (b) Give a careful proof of the Gershgorin Circle Theorem.
- (c) Let

$$A = \begin{pmatrix} 5 & .01 & .03 \\ .02 & 4 & .01 \\ .03 & -.01 & 4 \end{pmatrix}.$$

What can you say about the location of the eigenvalues of A ?

Part II: Numerical Analysis

6. This problem has three parts.

- (a) State Newton's method (the algorithm) for solving $f(x) = 0$, where $f : \mathbb{R} \rightarrow \mathbb{R}$.
- (b) Assuming f is smooth and we have an initial guess sufficiently close to the root, state sufficient condition(s) for the method to converge quadratically.
- (c) How would you use Newton's method to compute $5^{1/3}$?

7. This problem has three parts.

- (a) Give an exact expression for a Lagrange polynomial of degree n which would interpolate $n + 1$ given data points, $f(x_k)$.
- (b) Describe the differences between Hermite polynomials and Lagrange polynomials.
- (c) Suppose we have a set of Lagrange polynomials of degree $2n + 1$ which interpolate a smooth function at the points $x_0, x_0 + h, x_1, x_1 + h, x_2, x_2 + h, \dots, x_n, x_n + h$. What is the pointwise limit of this interpolant as $h \rightarrow 0$?

8. This problem has two parts.

- (a) Describe Romberg's technique for numerical approximation of an integral (based on the trapezoid rule).
- (b) Give a brief justification of the method.

9. Triple Recursion Formula. If $\{\phi_n(x)\}$ is an orthogonal family of polynomials on $[a, b]$, with weight function $w(x) \geq 0$, and $n \geq 1$, then show that

$$\phi_{n+1}(x) = (a_n x + b_n)\phi_n(x) - c_n \phi_{n-1}(x),$$

for some constants a_n, b_n , and c_n . (Hint: Let $g(x) = \phi_{n+1}(x) - a_n x \phi_n(x)$, where a_n is a constant chosen so that $g(x)$ has degree n .)

10. Let $\{\phi_k\}$ be a system of orthogonal polynomials, with respect to a non-negative weight $w(x)$, on $[a, b]$. Show that each of the polynomials $\phi_k(x)$ must have exactly k simple zeros in $[a, b]$.